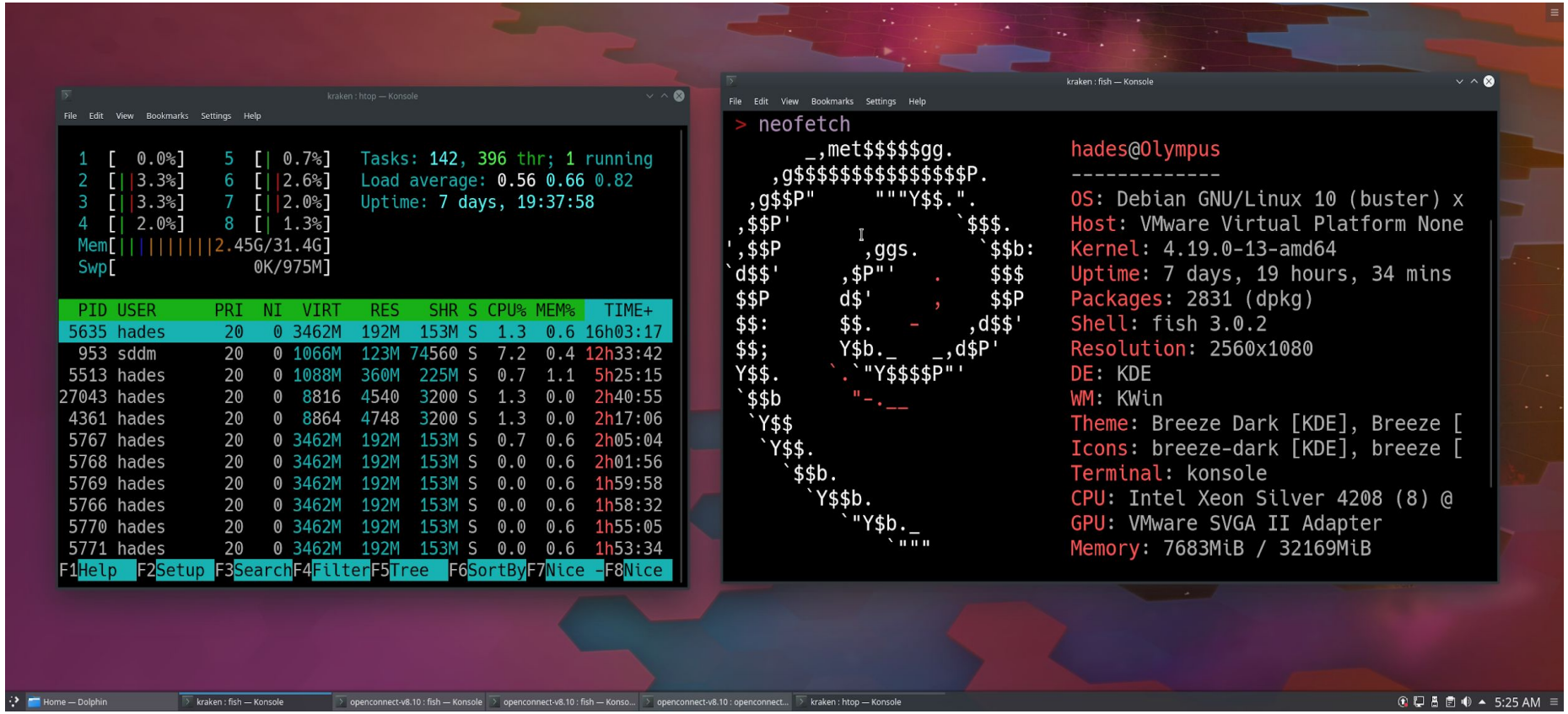
Rispro IEMS

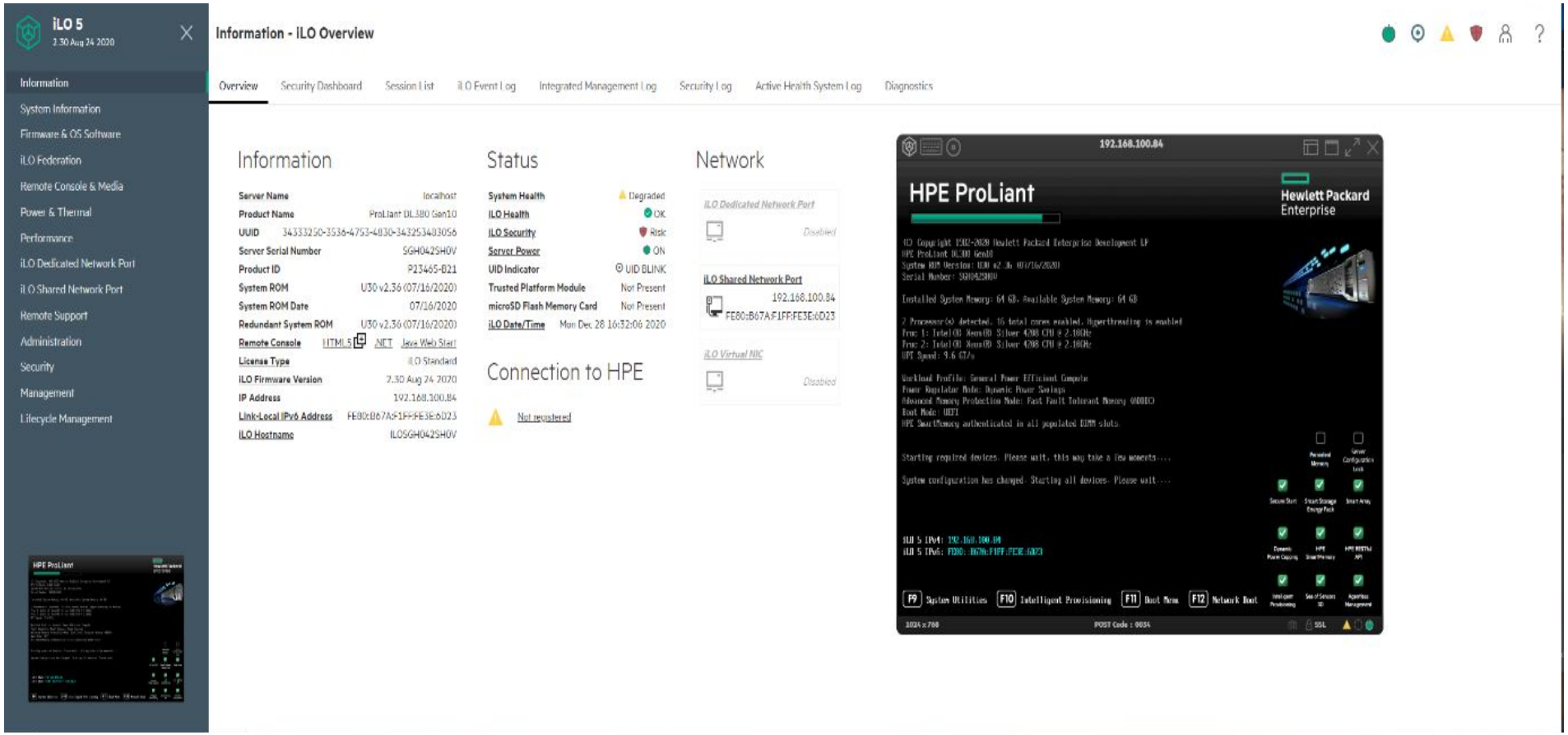
Server HPE Proliant Xeon Dual Processor

Gandeva Bayu Satrya and Team

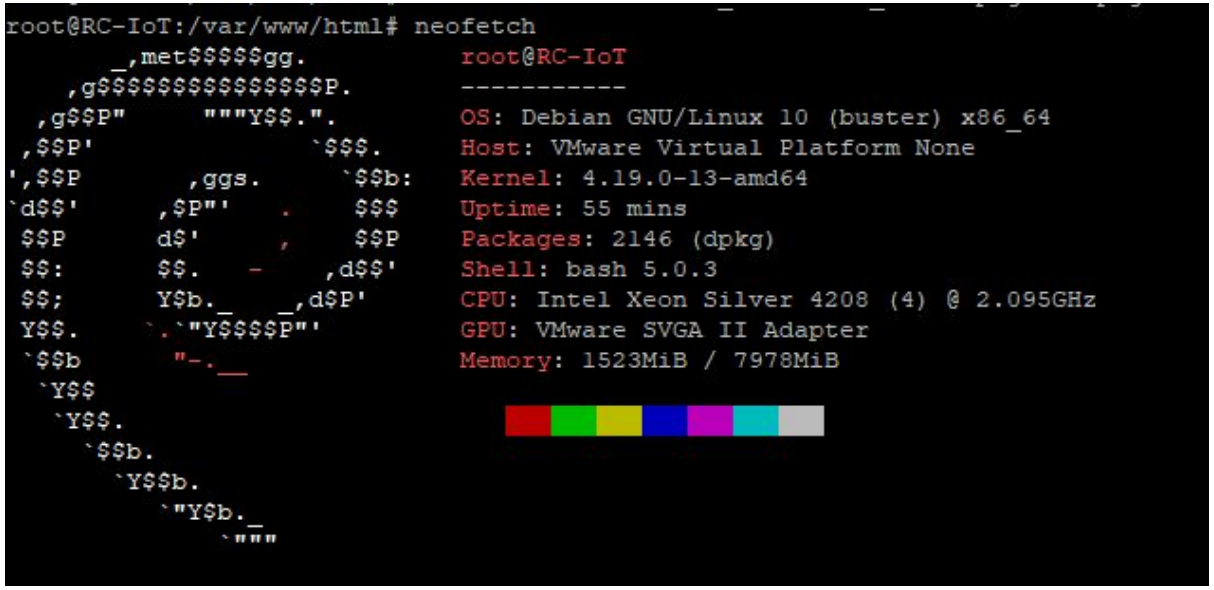
Pengujian Processor dan RAM

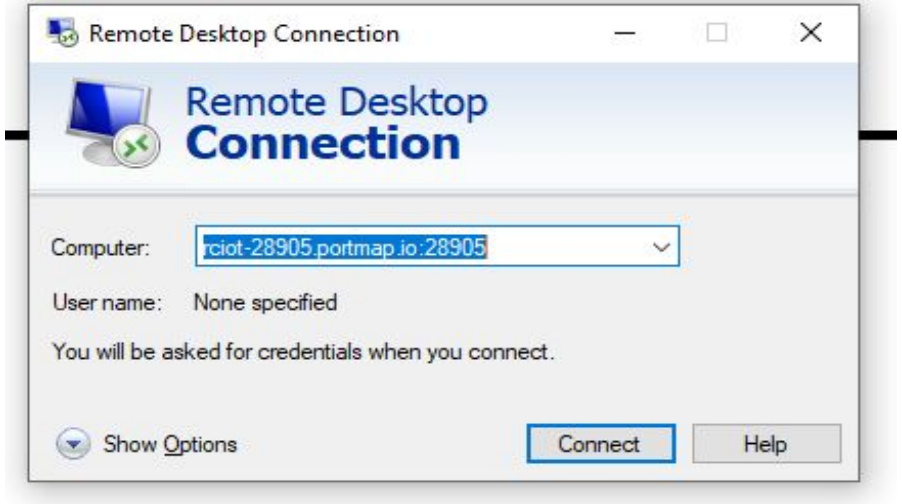


Instalasi HPE Proliant

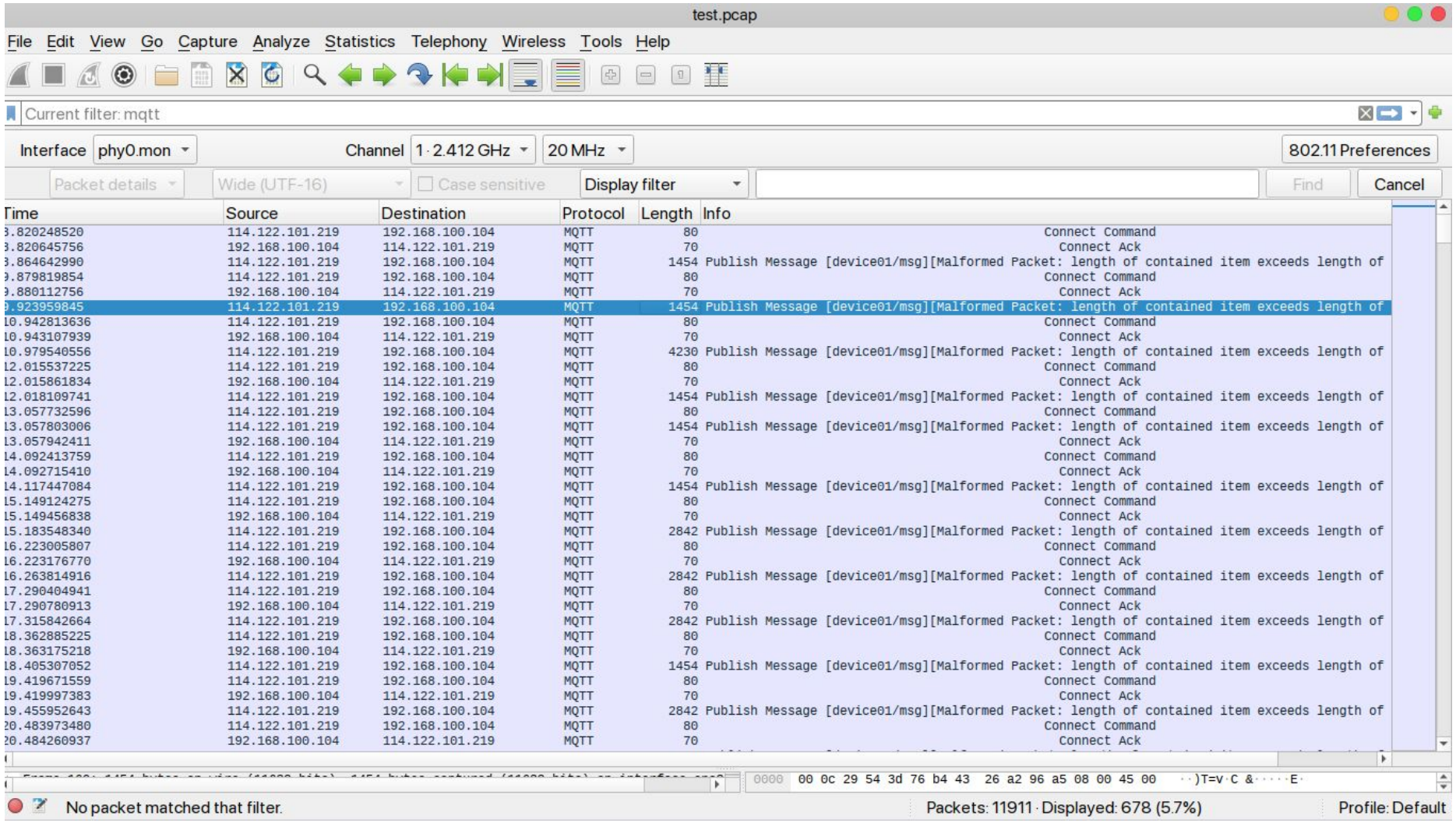


Pengujian Server Instance dari IP Public





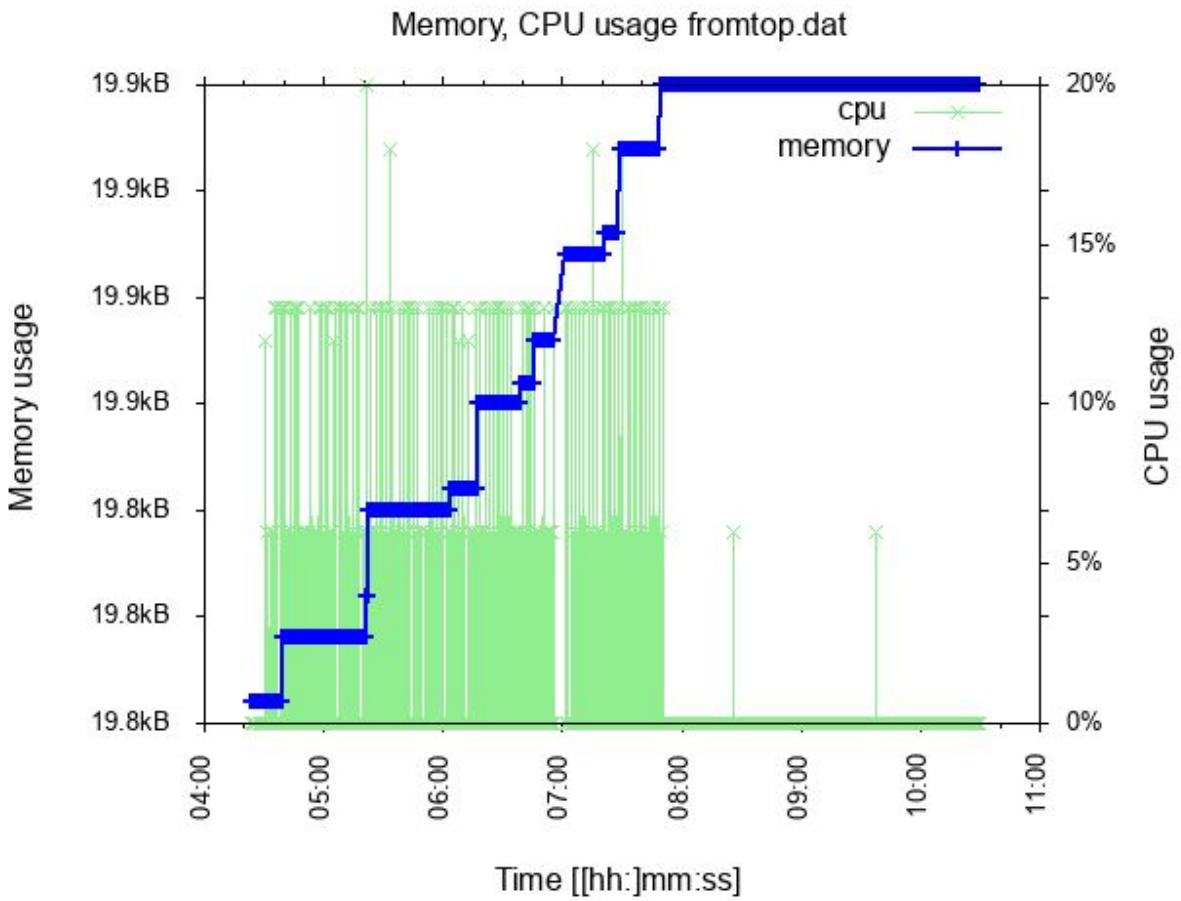
Hasil Pengujian data Terenkripsi



Time	Source	Destination	Protocol	Length	Info
3.820248520	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
3.820645756	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
3.864642990	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
9.879819854	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
9.880112756	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
9.923959845	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
10.942813636	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
10.943107939	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
10.979540556	114.122.101.219	192.168.100.104	MQTT	4230	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
12.015537225	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
12.015861834	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
12.018109741	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
13.057732596	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
13.057803006	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
13.057942411	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
14.092413759	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
14.092715410	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
14.117447084	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
15.149124275	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
15.149456838	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
15.183548340	114.122.101.219	192.168.100.104	MQTT	2842	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
16.223005807	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
16.223176770	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
16.263814916	114.122.101.219	192.168.100.104	MQTT	2842	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
17.290404941	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
17.290780913	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
17.315842664	114.122.101.219	192.168.100.104	MQTT	2842	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
18.362885225	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
18.363175218	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
18.405307052	114.122.101.219	192.168.100.104	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
19.419671559	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
19.419997383	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack
19.455952643	114.122.101.219	192.168.100.104	MQTT	2842	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
20.483973480	114.122.101.219	192.168.100.104	MQTT	80	Connect Command
20.484260937	192.168.100.104	114.122.101.219	MQTT	70	Connect Ack

0000 00 0c 29 54 3d 76 b4 43 26 a2 96 a5 08 00 45 00 ··)T=v·C &·····E·

Memory Usage and CPU Usage



Virtualisasi Server

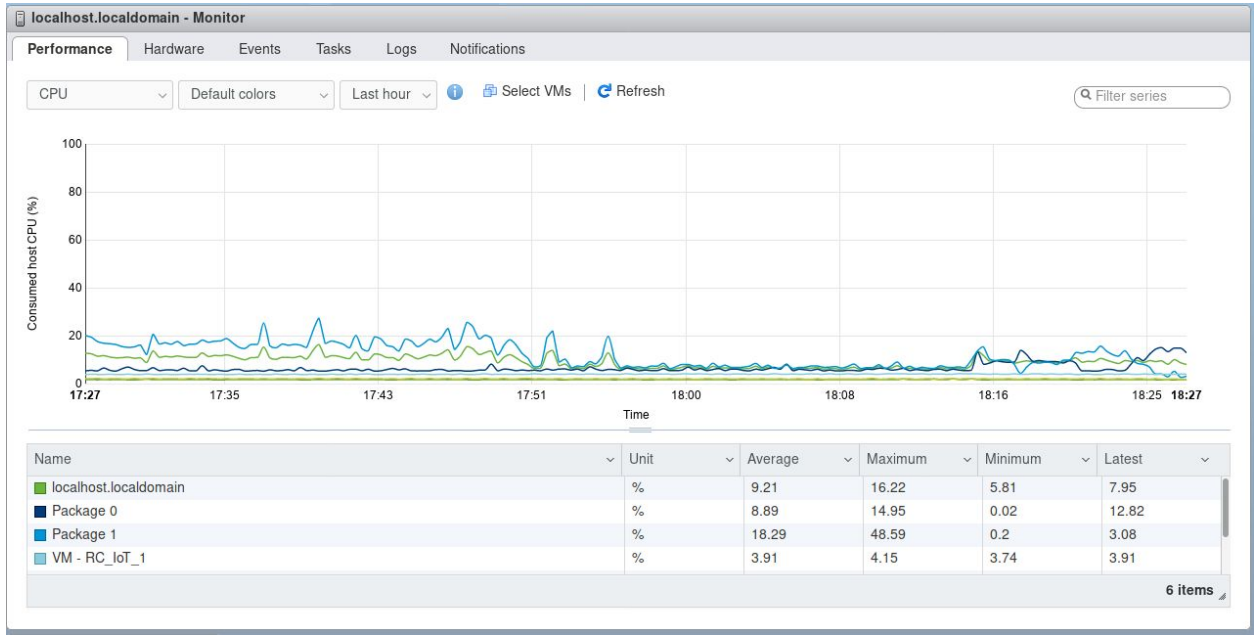
Create / Register VM | Console | Power on | Power off | Suspend | Refresh | Actions | linux

Virtual machine	Status	Used space	Guest OS	Tools i...	Host name	Host CPU	Host mem...
RC_IoT_1	Normal	108.09 GB	Debian GNU/Linux ...	No	Unknown	691 MHz	7.44 GB
Leviathan	Normal	50 GB	Debian GNU/Linux ...	No	Unknown	0 MHz	0 MB
RC_IoT_2	Normal	58.08 GB	Debian GNU/Linux ...	No	Unknown	72 MHz	990 MB
Uroboros	Normal	116.09 GB	Other Linux (64-bit)	No	Unknown	325 MHz	16.1 GB

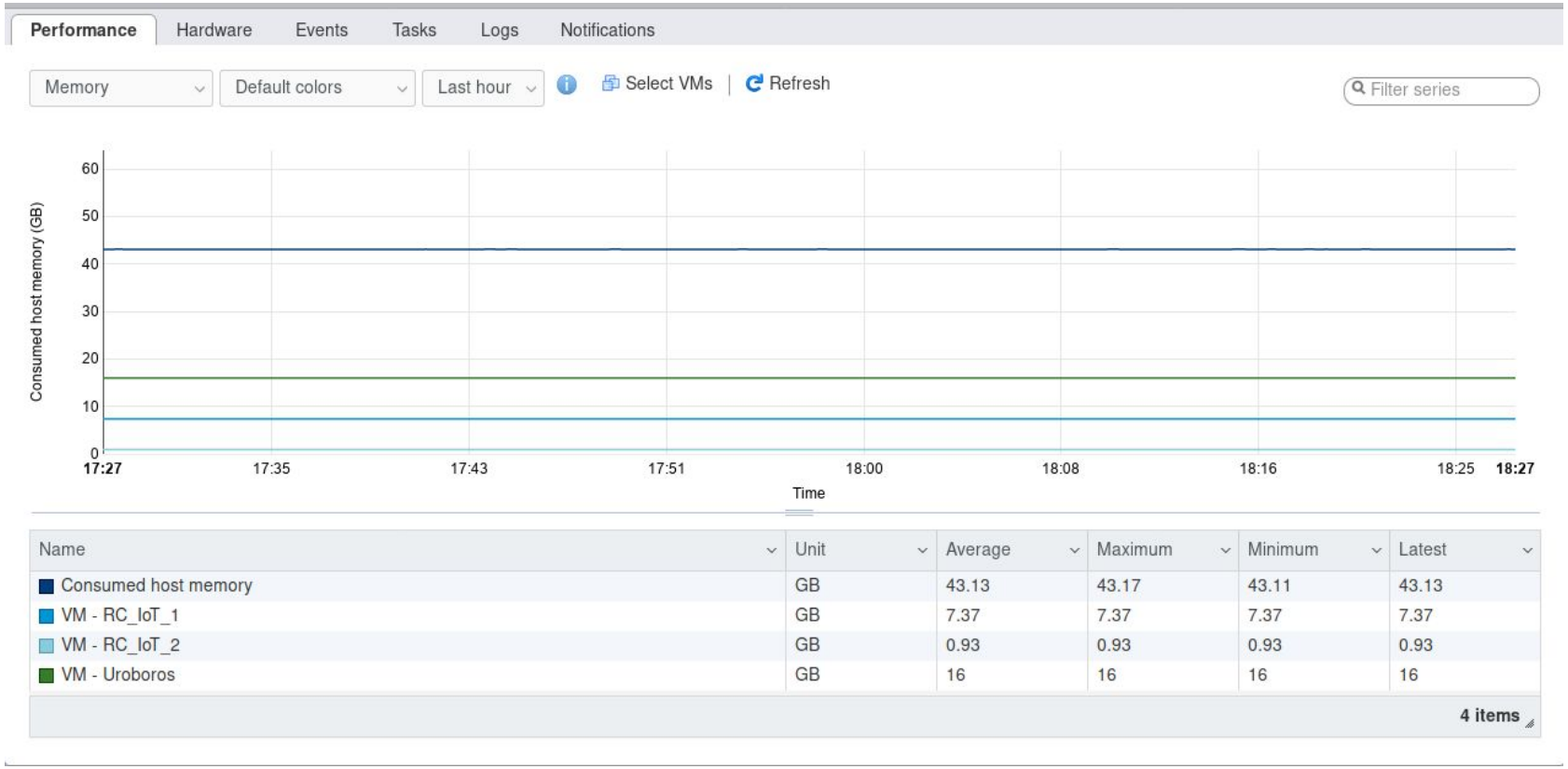
Quick filters...

5 items

Monitoring Penggunaan CPU Server



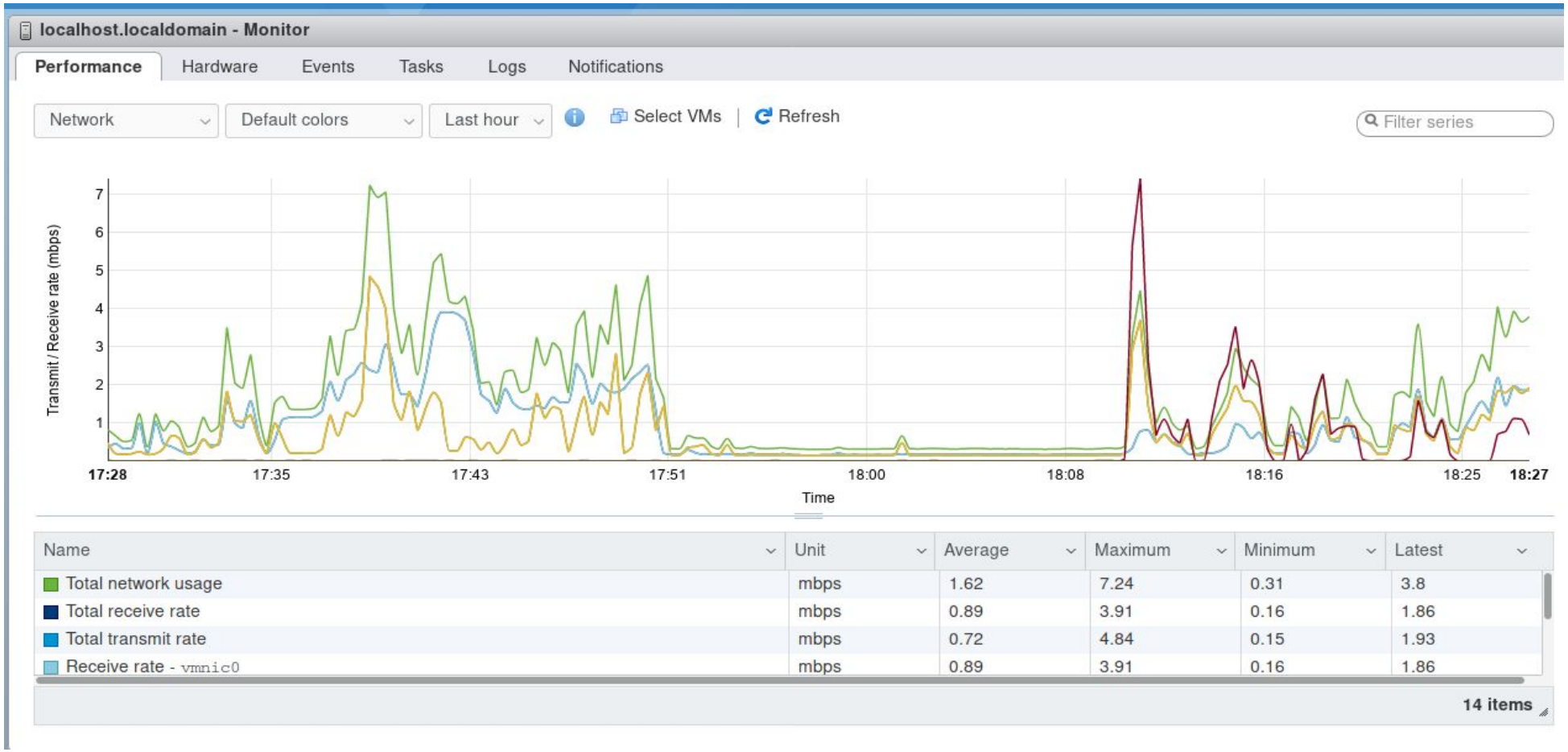
Monitoring Penggunaan RAM



Name	Unit	Average	Maximum	Minimum	Latest
Consumed host memory	GB	43.13	43.17	43.11	43.13
VM - RC_IoT_1	GB	7.37	7.37	7.37	7.37
VM - RC_IoT_2	GB	0.93	0.93	0.93	0.93
VM - Uroboros	GB	16	16	16	16

4 items

Monitoring Tx/Rx Pada Sisi Server

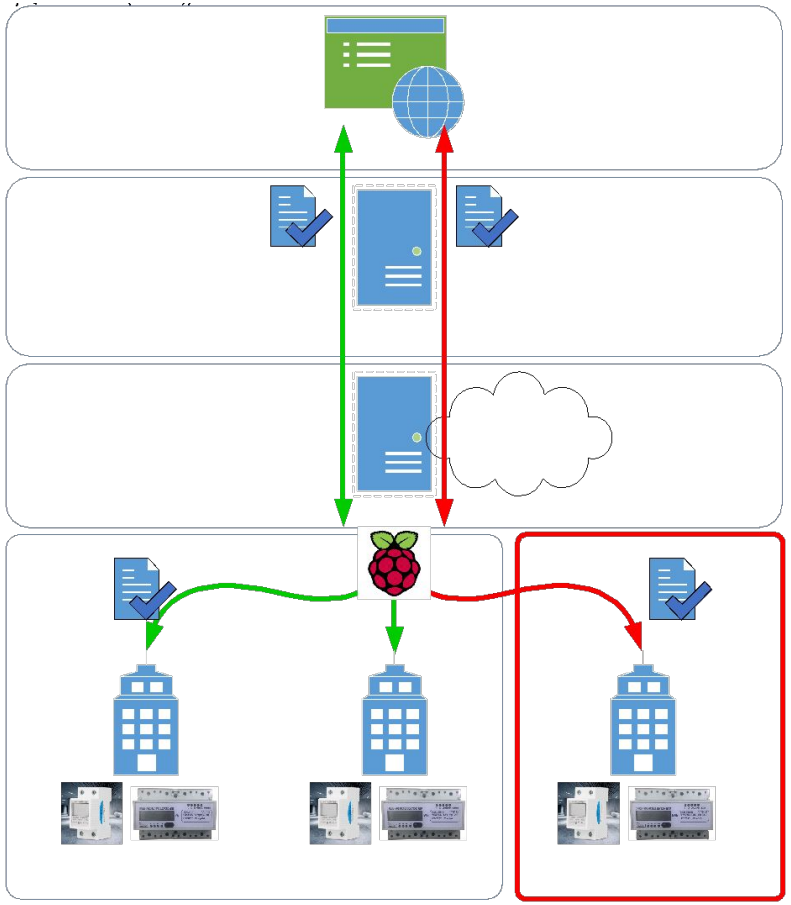


Rispro IEMS

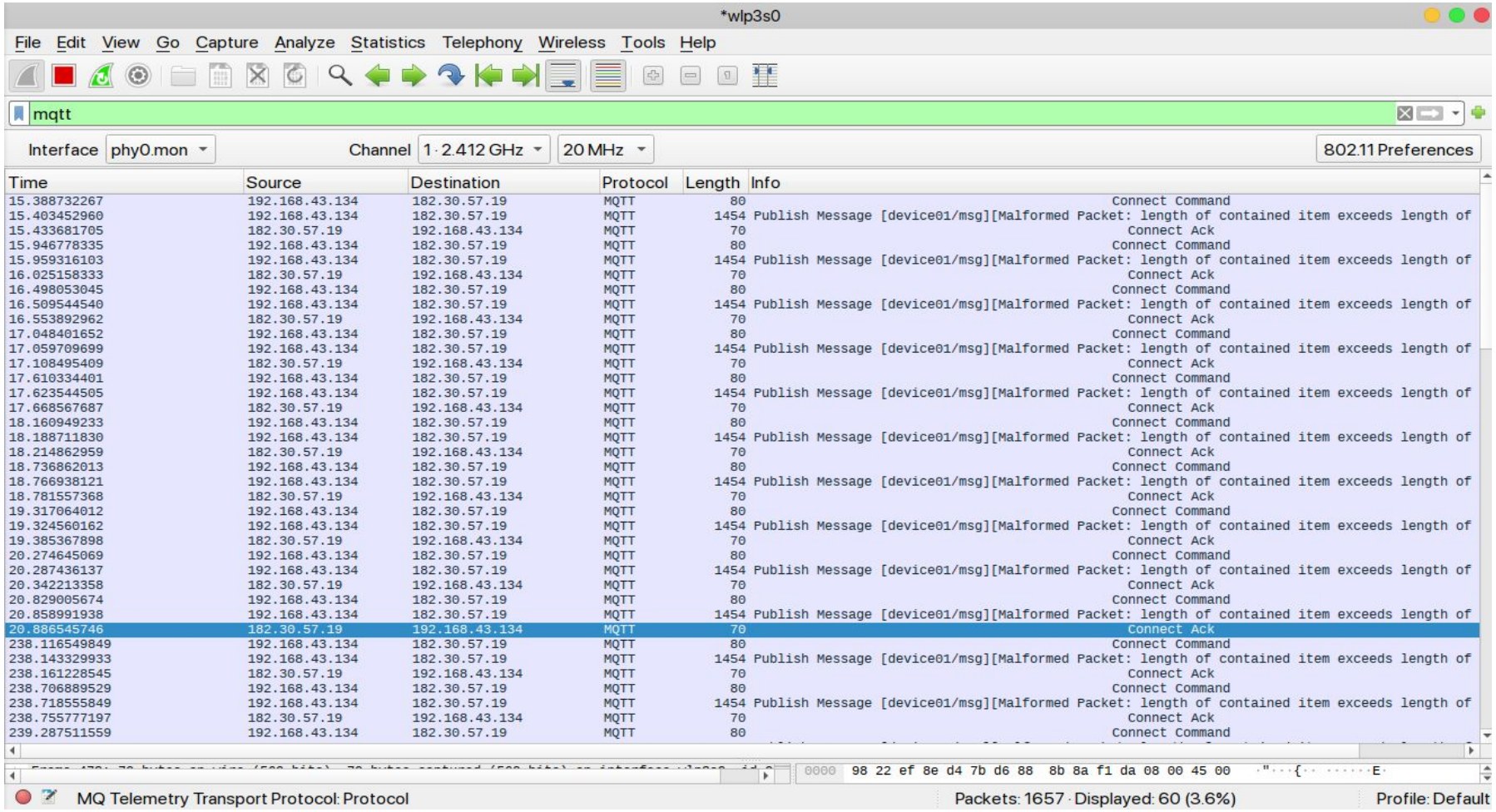
Secured Communication

Gandeva Bayu Satrya and Team

Implementation NTRU (Post-Quantum Cryptography) on Raspberry



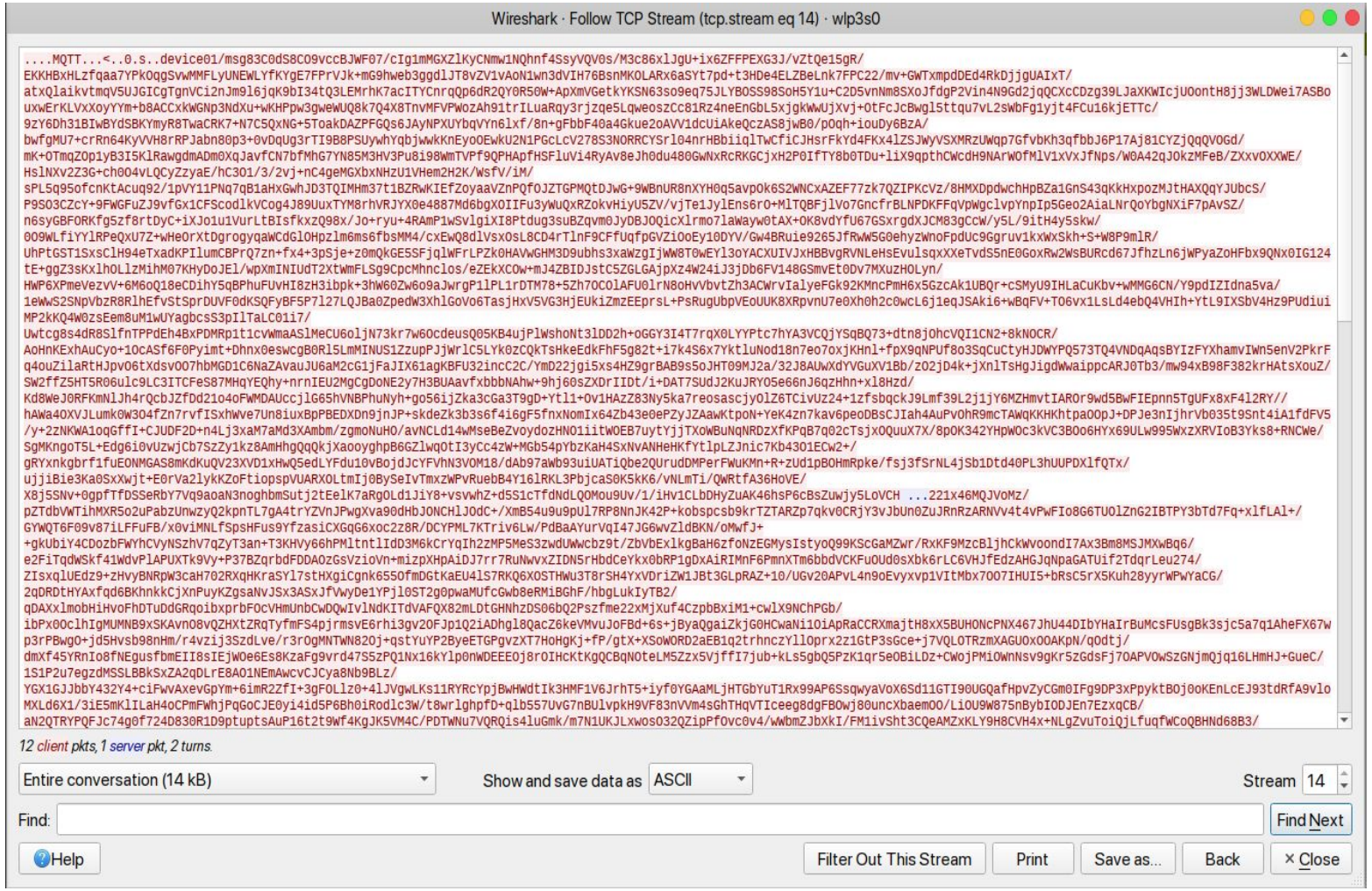
Secured Communication (Pub/Sub)



Time	Source	Destination	Protocol	Length	Info
15.388732267	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
15.403452960	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
15.433681705	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
15.946778335	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
15.959316103	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
16.025158333	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
16.498053045	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
16.509544540	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
16.553892962	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
17.048401652	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
17.059709699	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
17.108495409	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
17.610334401	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
17.623544505	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
17.668567687	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
18.160949233	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
18.188711830	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
18.214862959	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
18.736862013	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
18.766938121	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
18.781557368	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
19.317064012	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
19.324560162	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
19.385367898	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
20.274645069	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
20.287436137	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
20.342213358	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
20.829005674	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
20.858991938	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
20.886545746	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
238.116549849	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
238.143329933	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
238.161228545	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
238.706889529	192.168.43.134	182.30.57.19	MQTT	80	Connect Command
238.718555849	192.168.43.134	182.30.57.19	MQTT	1454	Publish Message [device01/msg][Malformed Packet: length of contained item exceeds length of
238.755777197	182.30.57.19	192.168.43.134	MQTT	70	Connect Ack
239.287511559	192.168.43.134	182.30.57.19	MQTT	80	Connect Command

0000 98 22 ef 8e d4 7b d6 88 8b 8a f1 da 08 00 45 00

Pesan Terkirim (Encrypted by NTRU)



Encryption Process on Publisher

```
===== [ Sequence  15 ] ======================================================
File destination : ./cipher/cipher_EES401_07.dat
Python Log, C Return         :  0.0005280000041238964
Cipher Length                :  552

===== [ Sequence  16 ] ======================================================
File destination : ./cipher/cipher_EES401_07.dat
Python Log, C Return         :  0.0005549999768845737
Cipher Length                :  552

===== [ Sequence  17 ] ======================================================
File destination : ./cipher/cipher_EES401_07.dat
Python Log, C Return         :  0.0005300000193528831
Cipher Length                :  552

===== [ Sequence  18 ] ======================================================
File destination : ./cipher/cipher_EES401_07.dat
Python Log, C Return         :  0.00044100001105107367
Cipher Length                :  552

===== [ Sequence  19 ] ======================================================
File destination : ./cipher/cipher_EES401_07.dat
Python Log, C Return         :  0.00035700001171790063
Cipher Length                :  552

===== Result ================================================
Python LOG Plain Length              :  241
Python LOG Plain with Padding Length :  256
Sending Cipher with length           :  14720
Python LOG Plain Array               :  ["('230425', '2', ", "'234.88', ", "'235.68', ", "'234.76', ", "'407.51', ", "'407.41', ", "'406.72', '4.3'," , " '2.4', '6.5', ", "'246', '271', ", "'558', '0.95', ", "'0.43', '1.39', ", "'0.98', '0.5', ", "'1.49', '0.25', ", "'0.54', '0.37', ", "'50', '0', ", "'2507.63', ", "'2020-10-01 ", "15:35:50', ", "'NULL')", '                ']
 ~/P/P/M/C/NTRUEncrypt_Benchmark  master
 nino@pop-os  work@NTRUEncrypt_Benchmark        1:Vim-  2:Exec*  3:Debug  4:Files#  5:Doc#  6:Process#        15:13:18  2021-03-18
```

Receiving Messages on Subscriber

```
====[ Incomming Msg ]=======================
Incomming Message Length              :  14720
Before decryption : b'Kw\x13\x00\xbd\x84\xf0\xc1\x9dc\x1c\ta\xa0\x0b!9\x10?\xc0yF\x9e\xc7Y5\xa8\xc3\xa1\x9c\xb5\x94@\xb4\xc0\x9c.\x14\xc3(\x02\x96\xd
2\xe4\x1e\xacI\xe2\x0fz\x9a`w\xde\x032\xdf\xf2\xb2\x0f\xed7\x83\x86\xb4\x01\xfa\x9a\x08\xc4#\xa7Wld\xdb\x9e\xea\x02\x1e)\xa1\xd5"\xdc \xd9XG>\x85\xa5
\xddX|\xa2!\xde\xeb\x8d'...
Decryption Result (Plaintext)         :  ('230422', '2', '234.91', '235.83', '234.8', '407.67', '407.57', '406.78', '4.3', '2.7', '6.5', '251', '321',
 '553', '0.94', '0.49', '1.37', '0.98', '0.58', '1.47', '0.25', '0.55', '0.38', '50', '0', '2507.57', '2020-10-01 15:33:19', 'NULL')

====[ Incomming Msg ]=======================
Incomming Message Length              :  14720
Before decryption : b'9\x10\xe0\xc1x\x9d\xc0\x88\xfa\xf4\xd0\x90J\xdee\xf6\x956\xda\xb8\x81q\xb5\xdb;3\xbe\xc8\x91h\x05u\x08AB\x85\xf7\xd0\xefA\x02\x
d2\xf1\x0ed\xa0$VHX>\xf4\xce\xbfP\xda\xe8:\x0f@\xbep(\xe5a\xf1\xe8\xb3 \xf8\xf6\x11\xf7\x9ay\xa3}\xba\xcb^g\xfcb\x1cL\xfe\x12\xf7\x864\xb1M\x84\x8e}q
\x94\x9f\xdf\xbe'...
Decryption Result (Plaintext)         :  ('230423', '2', '234.91', '235.8', '234.69', '407.65', '407.46', '406.69', '4.8', '2.7', '6.7', '296', '326',
 '578', '1.08', '0.49', '1.43', '1.11', '0.59', '1.54', '0.27', '0.56', '0.38', '50', '0', '2507.59', '2020-10-01 15:34:09', 'NULL')

====[ Incomming Msg ]=======================
Incomming Message Length              :  14720
Before decryption : b"\xc0S\x84\xea\x94G\xee\xefb?E\x85\x11\xfc\xff\n\xc0s\xfc\x84\x88\xd4-\xe7D\x840\xfdd \x07\xa2\xc0\x0f\x02\xa8;\xcf\xdf\\\x99\x1
fj\xd8\x04=\x84Z\xc8\xa7\xd8\xc6\xeb\xad\x9a\xdby\x1d\x00\xf3\xd29\xc70\xb1\xf68\x07\x93\x8e$D\xfd\xc5^\xfaZ\x15\xa7\xbb\xf4\x99\xadw1\x9fG\x19'\x18H
x\xbc\xa2$\x1b\x17\xc9\xd9\xc9"...
Decryption Result (Plaintext)         :  ('230424', '2', '234.8', '235.67', '234.68', '407.44', '407.34', '406.58', '4.2', '2.7', '6.7', '246', '326',
 '583', '0.94', '0.49', '1.43', '0.97', '0.59', '1.54', '0.25', '0.56', '0.38', '50', '0', '2507.61', '2020-10-01 15:34:59', 'NULL')

====[ Incomming Msg ]=======================
Incomming Message Length              :  14720
Before decryption : b'_\xde\xbb\xeb\xa4\xf2\xbb\xde\xca\x85P\xdd\xaa;\x06/\xb6\x11\xdbc\xf4a\x99e\xac\x96\xa2\xdd\x04\xd5$\x90\x8b\xc17\xbc\xb4<}\xfd
\xa1a\xac=}\xe1\xa9\xd9\xeb\x8d\xd8\xbc\xd0y;@i2k1\\\x86\xe2&\xcb\xa5\xfa\xb9\x1fpM7\xcaJ|R\x8b_\xda\xc1[\xd2\xa4\xb4\xb2\x96\x06\xe8?\x0f0\x1d\x87-\
xf5w\xfc\xbc\x11\x97'...
Decryption Result (Plaintext)         :  ('230425', '2', '234.88', '235.68', '234.76', '407.51', '407.41', '406.72', '4.3', '2.4', '6.5', '246', '271'
, '558', '0.95', '0.43', '1.39', '0.98', '0.5', '1.49', '0.25', '0.54', '0.37', '50', '0', '2507.63', '2020-10-01 15:35:50', 'NULL')

hades@narrowmaze  18        1:python3*  2:vim-        15:13:53  2021-03-18
```

Encryption Time:
Comparison Result (NTRU, RSA, and Fernet)

NTRU	RSA	Fernet
0.015162	0.022758	0.00159
0.018289	0.004106	0.00091
0.018467	0.003689	0.00089
0.018904	0.003627	0.00090
0.018462	0.003931	0.00091
0.018319	0.003610	0.00099
0.018287	0.003601	0.00090
0.018468	0.003662	0.00091
0.017427	0.003622	0.00089
0.018411	0.003769	0.00091